

**Report
Of
Data Analyst – Agribusiness Data Analysis and Virtualization**

Submitted

To

**School of Computer Science and Engineering
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In partial fulfilment of the requirement of degree of

B.Tech CSE



By

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CANDIDATE'S DECLARATION

I, **PRACHI SINGH**, do hereby declare that the internship report titled “**Agribusiness Data Analysis and Virtualization**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

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ACKNOWLEDGEMENT

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I express my gratitude to my parents for their blessings.

Signature: _____

Student Name: PRACHI SINGH

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INTERNSHIP COMPLETION CERTIFICATE



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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report documents the work completed during a four-week Data Analyst virtual internship in the agribusiness domain, undertaken with Codec Technologies as part of the requirements for the degree of Bachelor of Technology in Computer Science and Engineering at IILM University. The internship was structured around a single, continuous analytics pipeline that begins with public agribusiness data and ends with benchmarked performance indicators and strategic recommendations, with each week building directly on the deliverable of the week before it.

Agribusiness Data Analysis — by farm cooperatives, regional planners, or agricultural investors — depends on timely, accurate, and well-communicated data across production, weather, and market dimensions. However, the underlying public data is scattered across multiple government and intergovernmental sources, is rarely analysis-ready, and is not, by itself, predictive or evaluative. This project was designed to demonstrate the full journey from raw public data to actionable business insight: collecting and cleaning the data (Week 1), visualizing it for stakeholder reporting (Week 2), forecasting future trends from it (Week 3), and evaluating performance against industry benchmarks to generate strategic recommendations (Week 4).

The remainder of this chapter presents the organization profile, the problem this project addresses, its objectives and scope, the technologies used, the system architecture, the detailed methodology followed across all four weeks, and the expected outcomes of the completed pipeline.

4.2 Organization Profile

Codec Technologies is an information technology company that provides technology solutions alongside structured training programs and virtual internships for students and early-career professionals. The organization offers remote, flexible internship tracks across multiple technical domains, including Artificial Intelligence, data science, software development, cybersecurity, and cloud services, allowing learners to participate from any location.

Internship programs at Codec Technologies are designed around hands-on, industry-style project work rather than purely theoretical instruction, with structured weekly tasks, mentorship, and a defined set of deliverables submitted at each stage. This Junior Data Analyst – Agribusiness Virtual Internship followed that same model: each of the four weeks specified a concrete deliverable building progressively toward a complete, end-to-end analytics capability by the end of the program.

4.3 Problem Statement

Public agribusiness data — crop production statistics, weather records, and commodity prices — is openly available from government and intergovernmental sources, but it is fragmented across multiple platforms, inconsistently formatted, and not directly usable for decision-making in its raw form. Agribusiness stakeholders who want to understand production trends, communicate performance to others, anticipate future conditions, or evaluate their operations against industry norms currently have no single, structured, low-cost workflow for turning this scattered public data into decision-ready insight.

This project addresses that gap by designing and, where feasible, implementing a complete analytics workflow — covering data collection and cleaning, visual reporting, predictive forecasting, and KPI-based performance evaluation — built entirely on free, publicly accessible agribusiness data sources.

4.4 Project Objectives

1. To identify and evaluate publicly accessible agribusiness data sources spanning production, weather, and market-price data, and to design a repeatable process for collecting, exploring, and cleaning this data.
2. To design and demonstrate a data visualization and reporting strategy that communicates key agribusiness trends clearly to both technical and non-technical stakeholders.
3. To design a predictive analytics framework capable of forecasting agribusiness trends such as crop yield and commodity price, including a defined approach to model validation and evaluation.

4. To define key performance indicators (KPIs) for agribusiness operations, benchmark them against industry reference values, and translate the findings into actionable strategic recommendations.
5. To consolidate all four stages into a single, coherent, reproducible analytics pipeline, and to implement a working demonstration of that pipeline in a Python-based Google Colab notebook.

4.5 Scope of the Project

4.5.1 In Scope

- Publicly available data sources only — USDA NASS, FAOSTAT, NOAA Climate Data Online, and World Bank commodity/market data.
- Conceptual and applied work across all four internship stages: data cleaning, visualization, forecasting, and KPI evaluation.
- A working, executable demonstration notebook (Google Colab) implementing the full pipeline on a realistic representative dataset.
- Crop-level agribusiness metrics (yield, production, price) at a regional level of granularity.

4.5.2 Out of Scope

- Proprietary, farm-level, or real-time operational data belonging to any specific commercial entity.
- Deployment of the pipeline as a live, production-grade software system or dashboard product.
- Execution of the predictive models on verified real-world data with contractual accuracy guarantees — Week 3 in particular was scoped as conceptual planning and validation-design, consistent with the internship's task brief.

4.6 Technologies and Tools Used

The project used a combination of public data APIs, Python-based data science libraries, and standard office/reporting tools, summarized in Table 4.1 and Table 4.2 below.

Table 4.1: Public Data Sources Used for the Project

Source	Domain	Data Provided
USDA NASS QuickStats	Production	County/state/national crop yields, planted and harvested acreage, livestock inventories
FAOSTAT (UN FAO)	Production/Trade	Global crop and livestock production, trade flows, land use, and food balance sheets
NOAA Climate Data Online	Weather/Climate	Daily/monthly precipitation, temperature, and drought indices by station/region
World Bank Commodity Markets	Market Prices	Monthly global commodity prices and macro agricultural indicators

Table 4.2: Technologies and Tools Used

Category	Tools / Technologies
Programming & Analysis	Python 3 — pandas, NumPy, scikit-learn, statsmodels
Visualization	matplotlib, seaborn; Power BI / Tableau and Microsoft Excel recommended for interactive dashboards
Execution Environment	Google Colab / Jupyter Notebook
Forecasting	Random Forest and Linear Regression (scikit-learn), ARIMA and Holt-Winters exponential smoothing (statsmodels)
Documentation & Reporting	Microsoft Word (docx), Google Colab notebook export

4.7 System Architecture

Figure 4.1 presents the overall system architecture that connects all four weeks of the internship into a single pipeline. Public data from four sources feeds into the Week 1 data collection and cleaning stage, whose analysis-ready output feeds Week 2 (visualization), Week 3 (forecasting), and, together with all prior outputs, Week 4 (KPI evaluation and recommendations). The architecture includes a continuous feedback loop, since real deployment of this pipeline would repeat every season as new data becomes available.

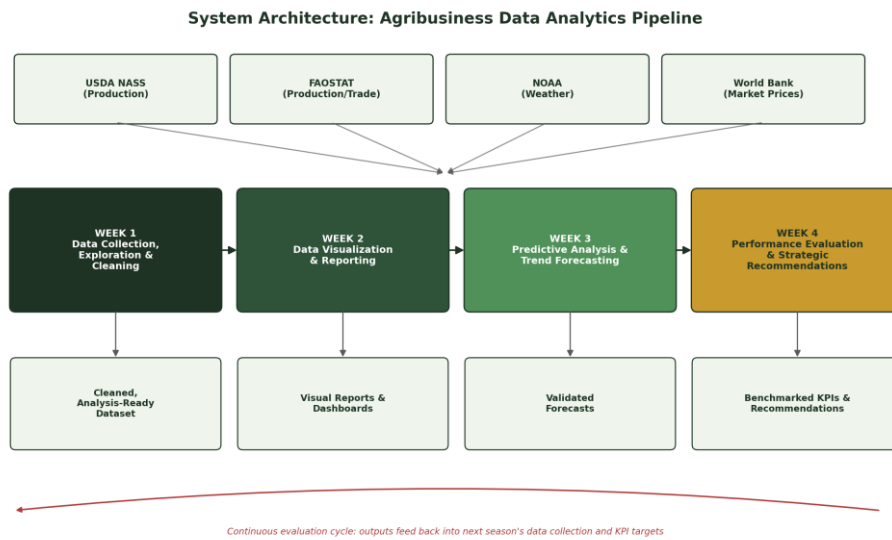


Figure 4.1: System Architecture of the Agribusiness Data Analytics Pipeline

4.8 Methodology

4.8.1 Week 1 — Data Collection, Exploration, and Cleaning

Data collection began with scoping the target commodities, geography, and time window, followed by extraction from each of the four public sources into separate raw files with recorded provenance metadata (source URL, retrieval date, and query parameters). Initial exploration computed null counts, summary statistics, and distribution plots for each numeric field, and checked that geography and year keys were consistent across sources before any merge was attempted.

Cleaning addressed three categories of data-quality issue. Missing values were quantified per column and imputed only for sparse gaps in continuous series (using regional means), while missing target values (yield) were never imputed. Normalization standardized units (a single yield unit, a common currency and base year for prices) and geographic/time keys (ISO or FIPS codes, four-digit years) so that all four sources could be merged reliably. Error correction applied agronomic range checks (yield cannot be negative; harvested area cannot exceed planted area), removed duplicate records using composite keys, and standardized inconsistent category labels through a lookup crosswalk. Figure 4.2 illustrates this cleaning sub-process.

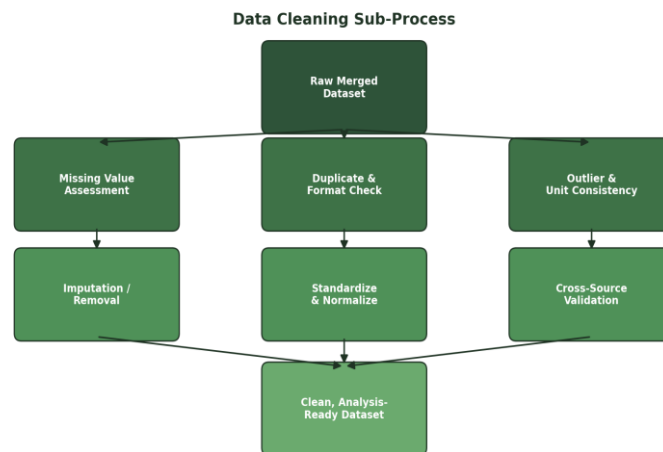


Figure 4.2: Data Cleaning Sub-Process

Every cleaning rule applied was logged with the affected column, number of rows changed, and rationale, so that the cleaned dataset could be reconstructed from the raw data plus the change log alone — a requirement for the dataset to be trustworthy input to Weeks 2–4.

4.8.2 Week 2 — Data Visualization and Reporting

Five categories of agribusiness metric were selected for visualization: production statistics, revenue/price trends, price volatility, regional comparisons, and weather–yield relationships. Each metric was matched to a specific chart type chosen for how well it communicates that kind of relationship — grouped bar charts for categorical production comparisons, line graphs for continuous price trends, scatter plots with a fitted trend line for the weather–yield correlation, and KPI summary cards for an executive-level overview. Figure 4.3 shows a representative sample chart from this stage, illustrating the color palette, labeling conventions, and overall visual style used throughout the reporting layer.

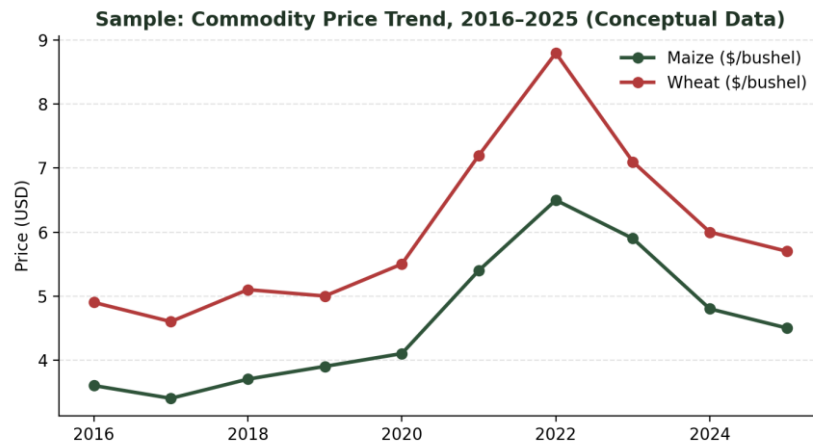


Figure 4.3: Sample Commodity Price Trend Visualization

The design process for this stage emphasized three principles: clarity (one primary message per chart, direct data labels, a consistent limited color palette), scalability (parameterized, script-driven charts that can absorb new years or regions without redesign), and aesthetic appeal (generous white space, muted gridlines, and a single consistent font family). The resulting report was storyboarded as a six-page sequence moving from an executive summary through production trends, regional comparison, price analysis, and weather correlation, to an appendix of data notes.

4.8.3 Week 3 — Predictive Analysis and Trend Forecasting

Three forecasting model families were evaluated for suitability: ARIMA (for single-series forecasts with limited external drivers), Holt-Winters exponential smoothing (for strongly seasonal series), and regression-based models (multiple linear regression and random forest regression, for forecasting driven by explanatory weather and market variables). A regression-based approach was selected as the primary method, since yield and price are influenced by identifiable external drivers (rainfall, temperature, prior-year values) rather than being explainable from their own history alone; exponential smoothing was retained as a simpler benchmark.

Table 4.4: Forecasting Model Comparison

Model	Best Fit
ARIMA	Single-series forecasts with a stable historical pattern and limited external drivers
Holt-Winters Exponential Smoothing	Strongly seasonal series (e.g., monthly commodity prices); simple to explain to non-technical stakeholders
Regression (Linear / Random Forest)	Forecasting driven by explanatory variables such as weather and market conditions; primary method selected for this project

Because agribusiness data is a time series, model validation used walk-forward (rolling-origin) validation rather than a random train/test split: the model trains on all data up to a cutoff season and is tested only on the single next season, after which the cutoff advances and the process repeats. This mirrors how the model would actually be used in production. Figure 4.4 illustrates this validation scheme.

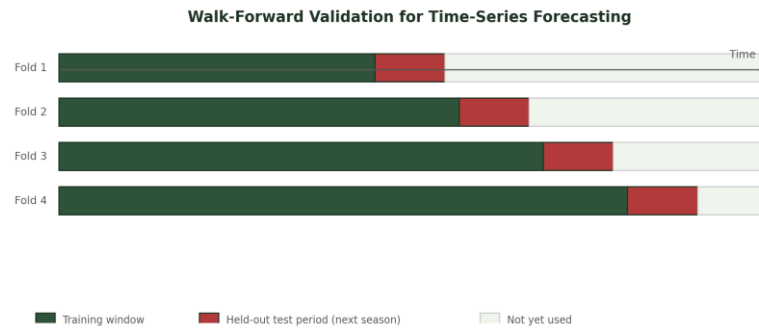


Figure 4.4: Walk-Forward Validation Scheme

Model performance was evaluated using four metrics — MAE and RMSE (in the original unit, with RMSE penalizing large misses more heavily), MAPE (for cross-commodity comparability), and R^2 (to confirm the explanatory variables are meaningful). In a hypothetical scenario, a regional grain cooperative uses the resulting yield forecast, together with its 95% confidence interval, to plan storage capacity and pre-negotiate forward contracts — sizing commitments toward the lower bound of the interval rather than the point estimate alone, illustrated in Figure 4.5.

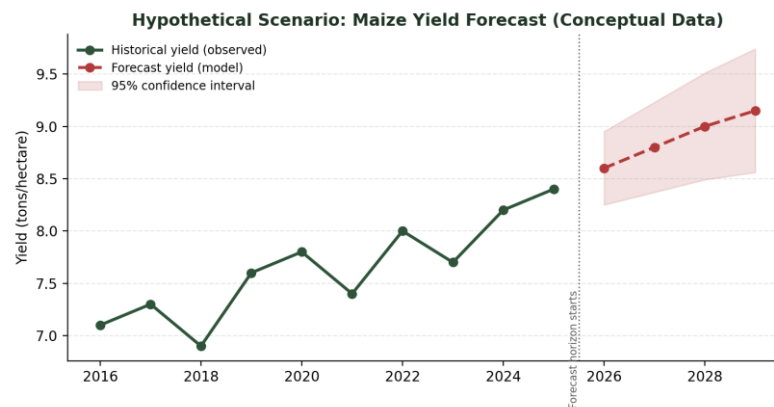


Figure 4.5: Hypothetical Yield Forecast with Confidence Interval

Five sources of bias and error were identified and mitigated by design: survivorship/reporting bias (mitigated by cross-checking regional coverage), structural breaks (mitigated by weighting recent seasons more heavily), overfitting (mitigated by walk-forward validation itself), omitted-variable bias (mitigated by explicitly stating

excluded drivers and communicating forecasts as ranges), and extrapolation risk in extreme years (mitigated by widening the confidence interval when inputs fall outside the training range).

4.8.4 Week 4 — Performance Evaluation and Strategic Recommendations

Five KPIs were defined to evaluate agribusiness performance across the full value chain, summarized in Table 4.3. Each KPI is normalized to a 0–100 performance score relative to a publicly sourced industry benchmark, so that KPIs measured in different units can be compared on one scale.

Table 4.3: Key Performance Indicators Evaluated

KPI	Category	Formula
Yield per Hectare	Production Efficiency	Total production (tons) ÷ harvested area (hectares)
Cost per Ton Produced	Cost Efficiency	Total cost ÷ total tons produced
Gross Profit Margin	Market Profitability	(Revenue – cost) ÷ revenue, as a percentage
Water Use Efficiency	Operational Sustainability	Tons of output ÷ total irrigation/effective water applied
Price Risk Coverage	Risk Management	Share of expected volume covered by forward contracts/hedges

A conceptual benchmark comparison (Figure 4.6) showed Yield per Hectare performing above benchmark, while Cost per Ton, Gross Profit Margin, Water Use Efficiency, and Price Risk Coverage all showed shortfalls, with Price Risk Coverage the largest gap. Gaps below 5 points are treated as normal variation, gaps of 5–15 points are flagged for a targeted improvement action, and gaps beyond 15 points are treated as priority findings requiring a strategic recommendation.

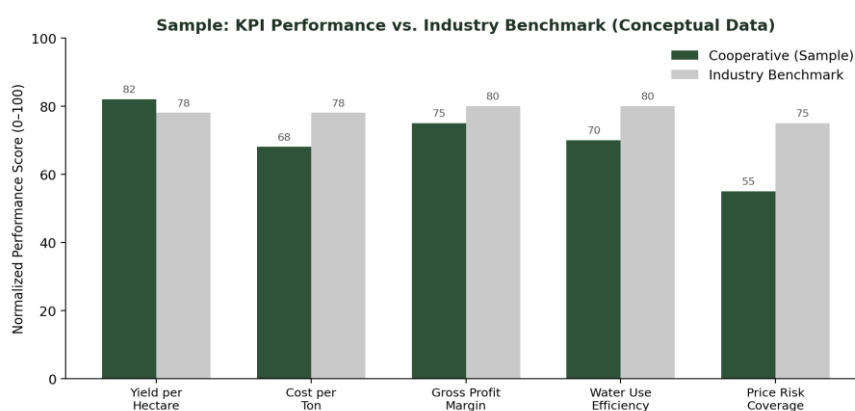


Figure 4.6: KPI Performance vs. Industry Benchmark

These findings translated into five strategic recommendations, summarized in Table 4.5: expanding forward contracting and price hedging (to close the largest benchmark gap); auditing input costs and piloting precision-application techniques (to close the cost-per-ton gap); introducing soil-moisture monitoring (to close the water-efficiency gap); protecting the existing yield advantage through continued agronomic investment; and treating gross margin as a downstream outcome metric that validates the other four actions rather than as a standalone initiative.

Table 4.5: Summary of Strategic Recommendations

Recommendation	Target KPI
Expand forward contracting and price hedging	Price Risk Coverage
Input-cost audit and precision-application pilot	Cost per Ton Produced
Soil-moisture monitoring for irrigation timing	Water Use Efficiency
Continued agronomic investment	Yield per Hectare (protect existing strength)
Treat margin as a downstream validation metric	Gross Profit Margin

Finally, ongoing evaluation was planned as a recurring Plan–Do–Check–Act cycle (Figure 4.7): set numeric KPI targets each season, collect data and compute KPIs on a consistent schedule, benchmark and classify the resulting gaps, and adjust strategy before the next cycle begins — with a quarterly cadence recommended for cost/risk KPIs and a seasonal cadence for yield/water-efficiency KPIs.

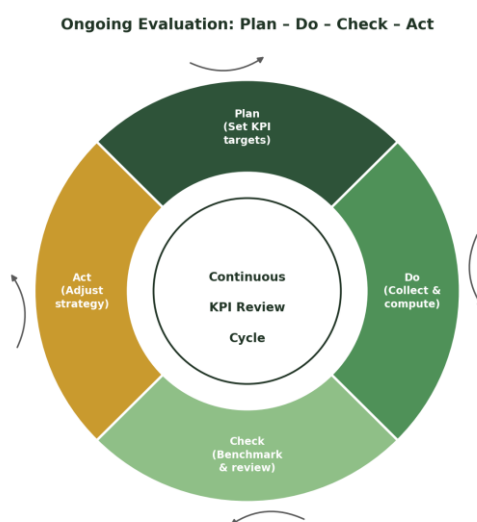


Figure 4.7: Continuous KPI Review Cycle (Plan–Do–Check–Act)

4.9 Expected Outcomes

- A cleaned, analysis-ready agribusiness dataset combining production, weather, and price data from four public sources, with a documented, auditable cleaning process.
- A structured visual reporting approach — chart-type justification, design system, and a six-page storyboard — ready to be populated with real data.
- A predictive forecasting framework, validated with walk-forward validation and benchmarked with four evaluation metrics, capable of producing yield and price forecasts with quantified confidence intervals.
- Five benchmarked KPIs and five corresponding strategic recommendations, along with a recurring Plan–Do–Check–Act evaluation cycle for ongoing performance management.
- A single, runnable Google Colab notebook implementing the entire pipeline end-to-end — from data generation/cleaning through visualization, forecasting, and KPI evaluation — executed successfully with zero runtime errors, and structured so that the synthetic data cells can be replaced with real public API calls.

Collectively, these outcomes demonstrate the ability to take a real-world business domain (agribusiness), work entirely from free public data, and produce a complete, professional, decision-ready analytics capability — the core skill set expected of a Junior Data Analyst.

4.10 Certificates of Completion and Other Communication Mail Proof



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